

Comparing More Than 2 Groups

How do we do statistical testing with more than 2 groups?

ANOVA!

One-Way ANOVA

Null Hypothesis

Mean of all the groups are the same.

Alternative Hypothesis

At least one of the groups has a mean that is different from the rest.

2 - Sample T-Test

Null Hypothesis

Mean of the two groups are the same.

Alternative Hypothesis

Mean of the two groups are not the same.

Demo

One Way ANOVA

Suppose we reject the null hypothesis based on ANOVA and conclude at least one of the groups was different. How do we figure out which one (post-hoc analysis)?

Pairwise Tests – Compare each combination of groups

Key Concepts

Familywise Error Rate:

Probability that at least one test results in a false positive (Type I Error) conclusion.

Bonferroni Correction:

Adjust the significance level to manage probability of a type I error when doing multiple tests.

Demo

Pairwise Tests

Suppose we wanted to compare flipper sizes for all penguins and all sexes

Gentoo Male	Gentoo Female
Chinstrap Male	Chinstrap Female
Adelie Male	Adelie Female

How do you feel about treating this as a One-Way ANOVA with 6 groups?

Suppose we wanted to compare flipper sizes for all penguins and all sexes

Gentoo Male	Gentoo Female
Chinstrap Male	Chinstrap Female
Adelie Male	Adelie Female

How do you feel about treating this as a One-Way ANOVA with 6 groups?

There's a better way!

What to do instead...

If you have 1 factor, do 1-Way ANOVA...

If you have 2 factors, do 2-Way ANOVA...

If you have 3 factors, do 3-Way ANOVA...

If you have X factors, do X -Way ANOVA

X-Way ANOVA

Null Hypotheses

- There is no difference in means in factor X
- There is no interaction between factors X, Y...

Alternative Hypotheses

- There is a difference in means in factor X
- There is an interaction between factors X, Y...

Demo

Two Way ANOVA



University of California
San Francisco